Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming 1st mid-term test– November 6th 2025

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and ensure to provide a brief explanation. Remember, without a reason, the answer will be considered invalid. Understanding the questions is crucial for your success in this exam.

A. Complete the following sentence and explain what it means: "A write conflict occurs when… (1)

B. Assuming T*(n) = T1(n), what kind of relation do you have between the efficiency and the speedup of a PRAM-based application? (1)

C. During the lectures, we discussed how the prefix sum and PRAM memory access enable us to maintain low algorithm complexity. Please discuss what type of PRAM memory accesses are performed. (1)

D. Please outline the key aspects of strong scalability (1)

Exercise n. 2

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Please explain how the "Select the *i* th smallest of *n* elements" can be solved (pseudo code required) and what kind of complexity we may expect from the reported solutions. (2)

B. Please explain how powering of a number works and how, with the Master Theorem method, you can compute its asymptotic complexity. (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

C. Describe how the union for Treaps data structure works and discuss its complexity. (1)

Exercise n. 3

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Why do we mention the BDD computed tables as an example of Dynamic programming? Please provide a pseudo-code example of its usage. (2)

B. Please describe what is peculiar to the self-organizing list problem, how this problem can be solved, and how its complexity could be analyzed. (1)

C. What properties and algorithmic steps are required to convert a divide-and-conquer solution into a dynamic programming solution? (1)

Exercise n. 4

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. What is the main improvement of the Karger–Stein minimum cut algorithm over Karger's original algorithm, and how does the complexity of one run compare to the overall complexity required to obtain the minimum cut with high probability in both cases? (2)

B. Please describe how the primality test, presented during the lectures, works, and specify which class of randomization it belongs to. (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**